



# Nested Loops in Python

## With Coding



# Outcomes of this Lecture

What is Nested Loop?

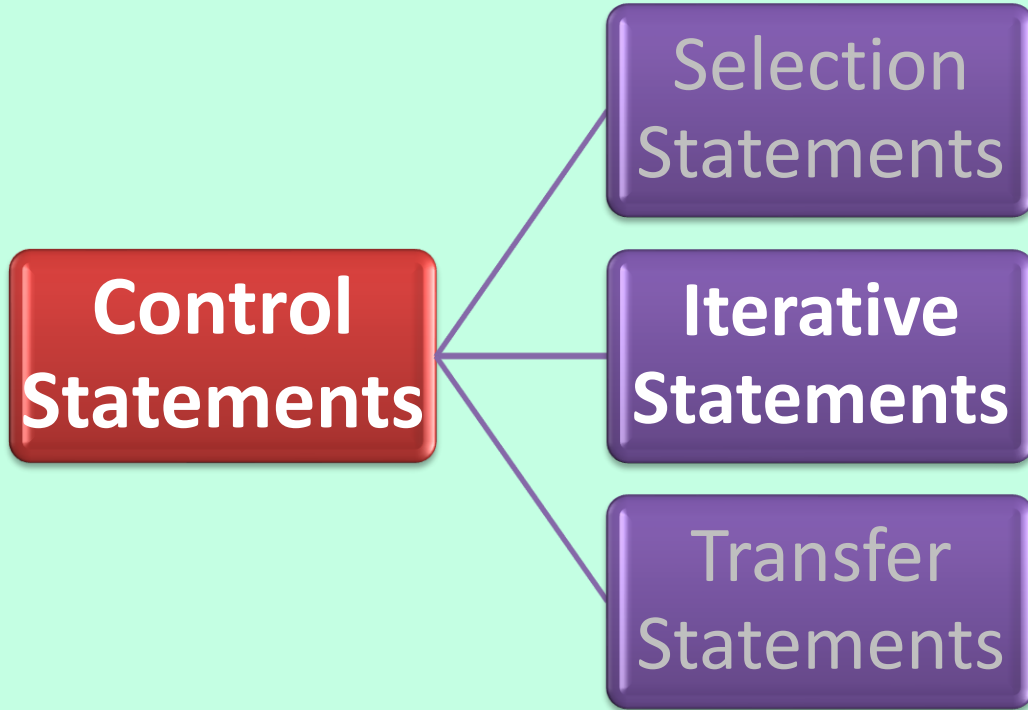
Execution procedure of Nested Loops

What is Nested for Loop?

What is Nested while Loop?



# Categories



# What is Nested Loop?



- In Python, a **nested loop** is simply a loop inside another loop. The "inner loop" will finish all of its iterations for every single iteration of the "outer loop."
- Think of it like a clock: the minute hand must go around **60 times** (inner loop) before the hour hand moves **once** (outer loop).



```
for outer_variable in outer_sequence:
```

```
    # Code in outer loop
```

```
    for inner_variable in inner_sequence:
```

```
        # Code in inner loop
```

```
    # This runs (outer count * inner count) times
```



# The Basic Logic

- When Python encounters a nested loop, it follows this flow:
  1. It enters the **outer loop** for its first item.
  2. It then enters the **inner loop** and runs it to completion (all iterations).
  3. It goes back to the outer loop for the second item.
  4. It repeats the inner loop again.

## OUTER LOOP (Rows, i)

Outer Loop: ``for i in range(2):`` (Run 2 times)

Start Outer Iteration (i=0)

Execute Outer Code (if any)

Start Outer Iteration (i=1)

END: Outer Loop Complete

## INNER LOOP (Columns, j)

Inner Loop: ``for j in range(3):`` (Run 3 times)

Enter Inner Loop

Iteration j=0 (Code runs)

Iteration j=1 (Code runs)

Iteration j=2 (Code runs)

End Inner Loop

```
for i in range(2):  
    print(f'Outer {i}')
```

```
    for j in range(3):  
        print(f'  Inner {j}')
```

|          |          |          |          |
|----------|----------|----------|----------|
| Outer 0, | Inner 0, | Inner 1, | Inner 2, |
| Outer 1, | Inner 0, | Inner 1, | Inner 2, |
| END      |          |          |          |



# When do we use Nested Loops?

This can be used

- **To work with Multi-dimensional data** such as Iterate over a matrix or **nested list**.
- ✓ Ex 1: To print elements of matrix by using row and column index.
- ✓ Ex 2: To print elements of the nested list (indexes of two lists).
- ✓ Ex 3: To work with multiple lists simultaneously.



# When do we use Nested Loops? Contd..

This can also be used

- ✓ **Generating Patterns** such as stars patterns, number patterns.
- ✓ Ex 1: To print stars/numbers in triangle shape.



# Nested For Loop

- We can write one for loop within the another for loop, is called Nested for loop.
- Syntax:

**Outer for Loop:**

**Inner for Loop:**

**Inner for loop Statement(s)**

**Outer for loop Statement(s)**



# Nested While Loop

- We can write one while loop within the another while loop, is called Nested while loop.
- Syntax:

Outer while Loop:

Inner while Loop:

Inner while loop Statement(s)

Outer while loop Statement(s)



## Points to remember

- Too many nested loops → code becomes slow and hard to read.
- Forgetting indentation → Python will throw errors.
- Infinite loops if using while without proper exit conditions.

# Check your knowledge

What is Nested Loop?

Execution procedure of Nested Loops

What is Nested for Loop?

What is Nested while Loop?

